北京郵電大學

Queen Mary University of London

For examiners' use only

1	
2	
3	
4	
Total	

EBU4201 A

Joint Programme Examinations 2018/19

EBU4201 Introductory Java Programming

Paper A

Time allowed 2 hours

Answer ALL questions

Complete the information below about yourself very carefully.

QM student number										
BUPT student number										
Class number										

NOT allowed: electronic calculators and electronic dictionaries.

INSTRUCTIONS

1. **You must NOT take answer books, used or unused, from the examination room.**
2. Write only with a black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Paula Fonseca, Dr Gareth Tyson, Dr John Woodward, Dr John Drake, Prof Simon Colton, Dr Ling Ma

Copyright © Beijing University of Posts and Telecommunications & © Queen Mary University of London 2019

Filename: 1819_EBU4201_A No answer book required

Introductory Java Programming EBU4201

1819A on QMPLUS

Solutions & Analysis

By Puyuan Zheng, BUPTIS.

(写在前面的话：解答题给分点均根据解析者本人对题目的理解认为的采分点。与实际若有不同，请以实际判分为准，本解析写出的采分点仅供参考。本解析若有问题错误，欢迎批评指正。)

Question 1: (25 Marks)

a) Suppose you are writing a Java program for an art gallery. (15 Marks)

i) Write a class called ArtWork which has two attributes: name (which is a String), and creation_year (which is an integer) indicating when the artwork was created. Add accessor (getter) and mutator (setter) methods for each attribute. Use these appropriately in a constructor for ArtWork; the constructor takes two variables as input, which correspond to the name and creation year of an artwork, and sets the name and creation_year attributes accordingly. (11 Marks)

【Answer】详见 Solution

【考点】类的书写

【解析】

根据题意，我们要求书写一个类。类里面包含两个 Attribute 和方法。同时对于 constructor 方法有参数的限定。根据题目信息一句句写代码。

Solution:

1、Write a class called ArtWork。

public class ArtWork{

} (1 Mark)

2、which has two attributes: name (which is a String), and creation_year (which is an integer) indicating when the artwork was created.

String name;

int creation_year; (3 Marks)

3、Add accessor (getter) and mutator (setter) methods for each attribute.

【Review】Accessor 和 Mutator（getter 和 setter 是啥）

简而言之，

getter 用来访问某变量，方法类型为被访问变量的类型，无参数；

setter 用来赋值设定某变量，方法类型为 void，有参数。

```
public xxx getter(){
    return variablename;
}
public void setter(xxx abc){
    this.abc=abc;
}
```

So,

Accessors:

```
public String getName(){
    return name;
}
public int getYear(){
    return creation_year;
}
```

Mutators:

```
public void setName(String Name){
    this.name=Name;
}
public void setYear(int Year){
    this.creation_year=Year;
}
```

(7 Marks)

4、Use these appropriately in a constructor for ArtWork; the constructor takes two variables as input, which correspond to the name and creation year of an artwork,

and sets the name and creation_year attributes accordingly.Use these appropriately in a constructor for ArtWork; the constructor takes two variables as input, which correspond to the name and creation year of an artwork, and sets the name and creation_year attributes accordingly.

接下来要求利用刚才的方法构建 constructor，input 要求两个，所以两个参数；之后设定名字和年份。那么方法里面放两个 setter 就 ok。

【Review】Constructor 构造方法的特点

无返回类型，void 也没有；方法和类名相同；默认为无参数；可以有多个

如果用户自定义了一个带参数的，默认无参数的就不可用。

如果用户自定义了多个，不管是否有参数均可使用。

So,

```
public ArtWork(String name, int year){
    setName(name);
    setYear(year);
} (11 Marks)  类名 1+参数 1+方法要求 2
```

Final Answer:

```
public class ArtWork{
    String name;
    int creation_year;
public String getName(){
    return name;
}
public int getYear(){
    return creation_year;
}
public void setName(String Name){
    this.name=Name;
}
public void setYear(int Year){
```

```
    this.creation_year=Year;
}
public ArtWork(String name, int year){
    setName(name);
    setYear(year);
    }
}
```

Finish. (11 Marks)

ii) Add a method called age to the ArtWork class in part i), which returns the age of an artwork object by subtracting its creation_year value from 2019. Briefly describe a problem with this method's specification, and a solution for the problem. (4 Marks)

【Answer】详见 Solution

【考点】类的书写

【解析】

根据题意，我们要求继续增加刚才的类，继续补充代码。要求 1、加入一个变量 age，要求是到从建馆到 2019 年计算。然后考虑这个方法的问题，并对此进行解决。（这个问题难想吗？。。。）

Solution:

本题只需要填写本题对应的代码即可。

还是根据题目信息进行作答。

1、Add a method called age to the ArtWork class in part i), which returns the age of an artwork object by subtracting its creation_year value from 2019.

```
public int age(int creation_year){
    return 2019-creation_year;
}
```

暂时能写出这些。

但是你们发现什么问题了吗？（接下来就问的这个）

如果这个 artwork 是 2020，2021 年建立的呢？肿莫办？！！

年龄显然不可能是负数吧，不合常理。

所以不能直接 return 2019-creation_year

如何避免？晚建立的当 0 岁处理（因为要返回年龄，要返回东西。那么只能这么做。类似宝宝没出生你说年龄是多少？没见过-1 岁的吧。。）并且提示用户该博物馆 2019 年还未建立，具体什么时间建立的。

（如果有其他合情合理的想法，也可以算作正确答案，总之大致就是这个意思）

2、Briefly describe a problem with this method's specification。

A problem with this method is that when creation_year is later than 2019, we could get negative age.

3、A solution for the problem:

方法里面修改一下。

```
if(creation_year>2019){
    System.out.println("This ArtWork in 2019 is not created, it is created in year "+creation_year);
    return 0;
}
else{
    return 2019-creation_year;
}
```

Final Answer:

A problem with this method is that when creation_year is later than 2019, we could get negative age.

A solution for the problem:

```
public int age(int creation_year){
    if(creation_year>2019){
```

```
        System.out.println("This ArtWork in 2019 is not created, it is created in year "+creation_year);
        return 0;
    }
    else{
        return 2019-creation_year;
    }
}
```

b) This question uses some of the code you wrote for class ArtWork in part a). Write the code for a tester class called ArtTester with a main method. In this class, an ArtWork object called "The Queen" from the year 1817 is constructed. The main method should also print out the age of the artwork. (4 Marks)

【Answer】详见 Solution

【考点】Tester 类的书写

【解析】

根据题意，我们要求写一个测试类，对输入某美术馆的数据进行测试。而且要求 print 年龄。（出来的年龄答案应该是 202）。2019-1817=202.

但我们要求的是写代码！！！！

Solution:

1、Write the code for a tester class called ArtTester with a main method.

```
public class ArtTester{
    public static void main(String[] args){
    }
}
```

2、In this class, an ArtWork object called "The Queen" from the year 1817 is constructed.

既然要使用 ArtWork 类，那么我们需要 ArtWork 实体。

先新建一个 object 然后对这个 object 进行操作。

新建 object 然后直接构建：

ArtWork awork= new ArtWork("The Queen", 1817);

3、The main method should also print out the age of the artwork.

最后输出年龄。

int age=awork.age(awork.creation_year);

System.out.println("The age of ArtWork "+ awork.name +" is "+age+".");

Also, 也可以直接一气呵成。

System.out.println("The age of ArtWork "+ awork.name +" is "+awork.age(awork.creation_year)+".");

Final Answer:

```
public class ArtTester{
    public static void main(String[] args){
        ArtWork awork= new ArtWork("The Queen", 1817);
        int age=awork.age(awork.creation_year);
        System.out.println("The age of ArtWork "+ awork.name +" is "+age+".");
    }
}
```

Finish. (4 Marks)

c) Answer the following questions about the Java programming language: (6 Marks)

i) What is the difference between x++ and ++x, in terms of their code behaviour? (2 marks)

【Answer】详见 Solution

【考点】Java basics - Assignments and Operators

【解析】

本题考查 x++和++x 的区别，从代码行为的角度说不同。这两个都是增量，是先增还是后增的问题。单独使用效果一样，那么放在具体的语境下呢？

Solution:

x++ is the post-increment, causes the variable x to use at current and incremented afterwards.

++x is the pre-increment, causes the variable x to incremented first, and then used in the current statement.

The difference is the use time of the variable. x++ uses x before increment, and ++x uses x after increment.

Finish. (2 Marks)

ii) Why would someone use a break statement inside a switch? (2 marks)

【Answer】详见 Solution

【考点】Java basics - The switch statements

【解析】

本题考查 switch 里面为什么要加 break？主要从 break 作用分析，之后说如果没有 break 会造成什么后果。

Solution:

Break statement inside a switch causes the remainder of the switch statement to be skipped. If the break statement is missed, all case will be executed since inside switch condition is just a boolean expression of variable, if the variable exists, all case are true in logic. (2 Marks)

iii) Briefly describe the reason for using the extends keyword.

【Answer】详见 Solution

【考点】Java basics - Inheritance

【解析】

本题考查使用 extends 这个关键词的原因。Extends 表示继承关系，那么带了 extends 就表明某类继承于其父类，可以拥有其父类的所有 attribute 和 method。不需要再重新调用写一遍。

Solution:

Extends keyword uses for subclass and superclass. With extends on subclass, they inherit the properties (attributes and operations) of their superclass and while using the method or attributes in superclass, we do not need to define in subclass again.

Question 2: (25 Marks)

a) Answer the following questions about arrays and ArrayLists:(7 Marks)

i) Indicate TWO differences between arrays and ArrayLists. (2 Marks)

【Answer】详见 Solution

【考点】Array / Arraylist 的区别

【解析】

根据题意，本题考查 Array 和 Arraylist 的区别。区别非常明显，课件里有提及。

Solution:

下面说的 3 点选 2 点说就行：

DIFFERENCE 1：关于数组大小，Array 必须事先确定，ArrayList 不用。

An array needs to know its size at time of creation, whereas an ArrayList does not.

DIFFERENCE 2：关于加入元素，Array 插入必须要说明确定的下标，ArrayList 可以不用知道下标。（但仅限往后按顺序插）

To assign an object in a regular array, you must assign it to a specific index. But assigning in an arrayList, it's not necessary to know the index.

DIFFERENCE 3：下标访问元素，Array 是中括号；ArrayList 是弧括号。

Arrays use array syntax ([ ]) to get the element, ArrayLists use standard dot notation to get the element inside with .get method.

ii) Write Java code to place the following set of Strings into both an array called someArray and an ArrayList called someAList: {"twenty", "ten", "one"}.

Note: Your code must include a declaration of all the relevant variables. (5 Marks)

【Answer】详见 Solution

【考点】Array / Arraylist 插入元素

【解析】

根据题意，本题要求写代码将元素插入进 Array 和 ArrayList 中。本题确定就插入 3 个元素，所以大小确定。注意定义数组大小以及插入方法在二者之间的不同。

Solution:

本题可以题目要求分开写，有 Code for array 和 Code for arraylist。

题目要求所有相关的变量全部必须定义，不能直接使用。

所以本题：

特别注意：既然要用 arraylist，必须要导入 java.util 库，否则用不了。

So，开头还要再加上 import java.util.*;

```
import java.util.*;
public class Insertarrayalist{ // Insert into an array and an arraylist.
    public static void main(String[] args){
    }
}
```

第一步，先定义数组和 3 个字符串变量

```
String[] someArray = new String[3];
```

```
ArrayList<String> someAlist = new ArrayList<String>();
String string1= "twenty";
String string2= "ten";
String string3= "one";
```

第二步，将三个字符串变量插入进数组以及 ArrayList。

插入数组：

```
someArray[0]=string1;
someArray[1]=string2;
someArray[2]=string3;
```

插入 ArrayList：

```
someAlist.add(string1);
someAlist.add(string2);
someAlist.add(string3);
```

Final Answer:

```
String string1= "twenty";
String string2= "ten";
String string3= "one";
```

Code for Array:

```
String[] someArray = new String[3];
someArray[0]=string1;
someArray[1]=string2;
someArray[2]=string3;
```

Code for ArrayList:

```
import java.util.*;
ArrayList<String> someAlist = new ArrayList<String>();
someAlist.add(string1);
someAlist.add(string2);
someAlist.add(string3);
```

Finish. (5 Marks)

（看答题区域的格式，直接”twenty”, “ten”, “one”也合理）。本题重点在于定义 arraylist 和 array 以及添加元素的方法。

ii)

Code for the *array*:

Code for the **ArrayList**:

b) This question is about access modifiers: (6 Marks)

i) private and public are two of the access modifiers that can be used in Java. Explain the meaning of access modification in Java, by comparing the private and public access modifiers, including any advantages or disadvantages. (2 Marks)

【Answer】详见 Solution

【考点】Access Modifiers

【解析】

根据题意，本题要求阐释 access modification 的含义，并且要通过对比 private 和 public 进行阐述，包括优缺点。

所以答案应该是含义+优缺点两部分，含义之中带着比较。

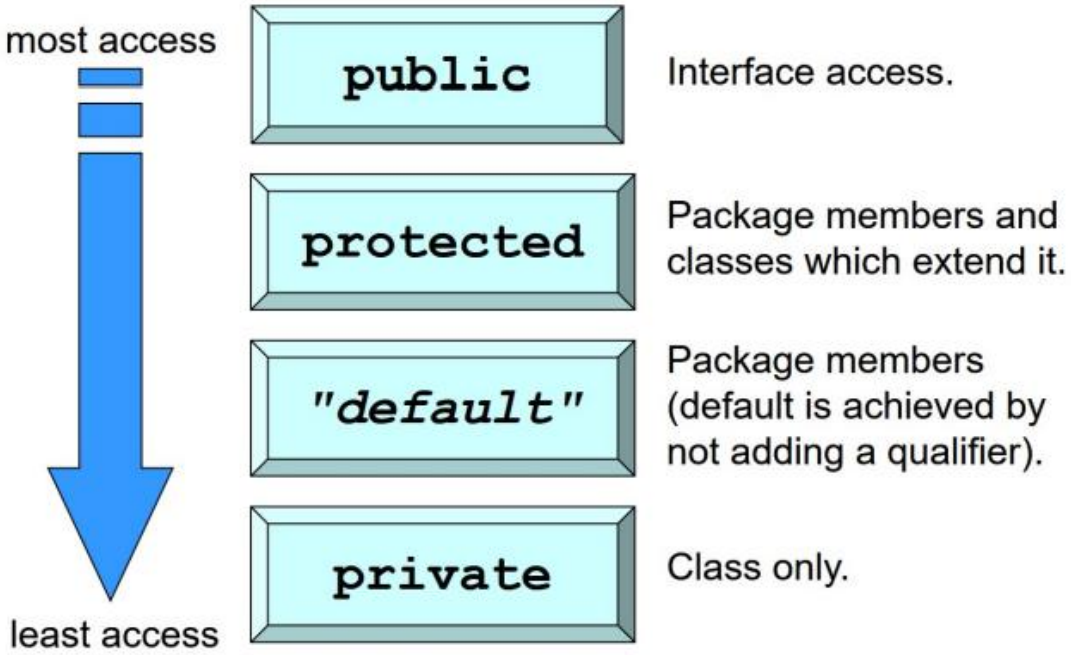
Solution:

Access modification is a kind of way to achieve information hiding. Public access modifier can let all information to be accessed in whole interface, but private access modifier only can be seen inside a certain class.

Since objects are only accessible through well defined interfaces, ideally nothing unexpected should happen, which achieve the purpose of object self-governing. Making internal change will not have impact on other classes. But, while constructing the code, for every variable and methods, we should consider the access modifiers to make sure the security of the code by following the relevant rules.

【Review】四个 Access Modifiers

Public， Protected， Default and Private



ii) How can you access an instance variable (attribute) if it is private? (2 Marks)

【Answer】详见 Solution

【考点】Access Modifiers

【解析】

根据题意，本题要求说明如何访问 private 变量。对于类内肯定可以直接访问，对于类外呢？

Solution:

In the same class, we can directly access it.

If we want an attribute to be accessible outside the class, we should do by accessor and mutator methods.

iii) Often we may store a fixed variable (i.e. a constant value) marked as public final static. For example, a constant such as the absolute zero value of temperature (i.e. -273.14 Kelvin) would be written as shown in Figure 1. Will making this public cause any problem? (2 Marks)

```
public final static double ABSOLUTE_ZERO = -273.14;
```

Figure 1

【Answer】详见 Solution

【考点】Numbers / Final methods

【解析】

根据图中给的常量，当这个常量是 public 会不会造成问题呢？对于正常的常量没什么问题，那么还有什么其他的情况？实际上是有的。对于数组！

Solution:

Yes. If public final static using on an array, since array is changeable, it causes security problem for code. (2 Marks)

Finish.

Reference:

https://wenku.baidu.com/view/a0113285b3717fd5360cba1aa8114431b80d8e45.html

c) Assume the class Flower has already been implemented with the correct constructors and methods (e.g. public String getName() and public String toString()). What does the code fragment (in Figure 2) output to the screen? (3 Marks)

```
Flower[] flowers = new Flower[3];
flowers[1] = new Flower("Rose");
System.out.println(flowers[1]);
System.out.println(flowers[2]);
System.out.println(flowers[3].getName());
```

Figure 2

【Answer】详见 Solution

【考点】Numbers / Final methods

【解析】

题意定义了一个 3 个大小的数组，但访问了下标 3，显然出现了异常。那么是什么？这是需要考虑的。此外，对于没有赋值的数组，访问了什么；对于 1，flowers[1]和 flowers[1].name 又有何区别？

Solution:

Flower@372f7a8d

null

Exception in thread "main" java.lang.ArrayIndexOutOfBoundsException: Index 3 out of bounds for length 3 at FlowerTest.main(FlowerTest.java:7)

（注意：flower[1]后面的地址编码考试中闭卷情况可以随便瞎写但要符合这样的格式；flower[2] 肯定是 null 毫无疑问；flower[3] 核心点在于 ArrayIndexOutOfBoundsException，Index 3 out of bounds for length 3，其他的个人认为次要。）开卷情况此题一定写代码 cmd 验证！

d) Given an array of int values, write a method that computes the maximum value. Hint: You should pay attention to the method header, return type, and the calculation itself. (6 Marks)

【Answer】详见 Solution

【考点】Arrays

【解析】

题意要求给一个数组，写一个找最大值的方法。要求返回最大值以及如何计算？对于数组找最大值通常采用刷新法。先设定最大值为[0]元素，往后遍历如果发现比当前最大值大那就覆盖。

Solution:

1、定义数组，但对于数组而言，需要事先知道有多大。但题目没有明确给出这一点，所以不考虑用 array。那么考虑 arraylist。

要用 arraylist 必须 import

```
import java.util.*;
```

2、之后定义方法：

```
public int Max(ArrayList<Integer> a){{
}
```

3、设定最大值变量 max

```
int max=a.get(0);
```

4、遍历刷新

```
for(int i=0;i<a.size();i++){
    if(a.get(i)>max){
        max=a.get(i);
    }
}
```

5、返回值

```
return max;
```

Final Answer:

```
public int Max(ArrayList<Integer> a){
    int max=a.get(0);
    for(int i=0;i<a.size();i++){
```

```
            if(a.get(i)>max){
                max=a.get(i);
            }
        }
        return max;
}
```

Finish. (6 Marks)

e) This section of the question is about Javadocs. (3 Marks)

【考点】javadoc

i) Write the command used to generate Javadoc comments, assuming that they are being generated for the file Example.java. (1 Mark)

【Answer】javadoc -d Example.java

【解析】

直接要求写出生成 javadoc 的指令。答案固定。而且要求是对 Example.java 生成。

Solution:

javadoc -d Example.java

【Review】javadoc 指令格式

javadoc -d 文档存放目录 -author -version 源文件名.java

文档存放目录当前文档或者没说可以不写，如果在不同文件夹下调该命令必须指明文件位置 -author -version 没说明或者不知道可以省略

ii) Give ONE example of a Javadoc tag, along with its description. (1 Mark)

【Answer】详见 Solution

【解析】

题目要求举一例 Javadoc 的标签，并且给出描述。主要有如下几个主要的：@author @param @return @throws @version，其他的不常见。

Solution:

只按上述提到的举例作答，其他的详见本题下参考链接：

@author author name, defines the author name of the class

@version info, tells the version information in this class.

@param parameter, tells the parameter a method use

@return returntype, tells the return type of a method

@throws exception, tells the exception possible to happen.

选一个就行。

更多的参考如下链接：

https://blog.csdn.net/languan1639397/article/details/122560483

iii) Explain what a Javadoc is, and what is its purpose. (1 Mark)

【Answer】详见 Solution

【解析】

题目问 javadoc 是什么以及其作用。固定答案。大体意思生成 Java API 文档，告知用户该代码的帮助文档。

Solution:

Javadoc is a tool which comes with JDK and it is **used for generating Java code documentation in HTML format from Java source code**, which requires documentation in a predefined format.

Reference: http://www.tutorialspoint.com/java/java_documentation.htm

(1 Mark)

Question 3: (25 Marks)

a) State whether each of the following statements about Java interfaces is TRUE or FALSE. (5 Marks)

i) An interface is a collection of abstract methods.

ii) The functionality of an interface is fully specified.

iii) It is possible for an interface to extend another interface.

iv) An interface cannot be instantiated.

v) A class can only implement a single interface.

【Answer】FALSE FALSE TRUE FALSE TRUE

【考点】Interface 基础概念

【解析】

题目要求对下列概念进行判断，要求对 interface 相关概念熟练掌握即可轻松作答。

Solution:

i) An interface is a collection of abstract methods.

Interface 里面可以不止抽象方法，Java SE8 之后又可以有 default 和 static method，**FALSE** 。

From Java SE8, **interfaces** can also have `default` and `static` methods. More about this, later in these slides …

ii) The functionality of an interface is fully specified.

FALSE

方法必须抽象 abstract，不能有任何内容，不能 specify。

iii) It is possible for an interface to extend another interface.

TRUE。相当于扩展接口的效果。

iv) An interface cannot be instantiated.

TRUE。和抽象类一样，不能被实例化。

v) A class can only implement a single interface.

一个类可以 implements 多个 interface。**FALSE**

Public class A implements B,C 没问题

但 extends 只能有一个类！！规定，否则就是 Dead Diamond 问题。

Finish. (5 Marks)

b) This question is about Java memory management: (10 Marks)

i) What are the TWO main areas of memory in Java? Give an example of what might be stored in each area. (4 Marks)

【Answer】Main areas: Heap and Stack

Example 详见 Solution

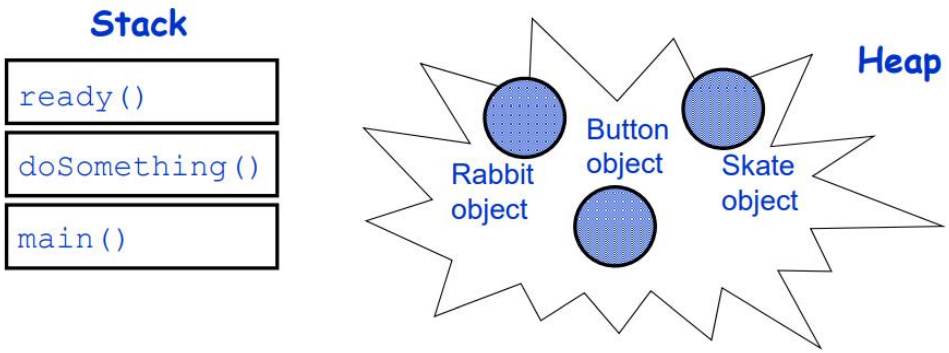
【考点】Memory in Java

【解析】

题目问 java 的空间存储在哪两个区域，死知识点。并且举例。

Solution:

- **Two main areas of memory in Java:**
  - (Garbage-collectible) Heap (where **objects** live);
  - Stack (where **local variables** and **methods,** when called, live).



TWO main areas of memory in Java: Heap and Stack. (2 Marks)

Objects are stored in heap. (3 Marks)

Local variables and methods which are called, are stored in stack. (4 Marks)

Finish.

ii) Assuming Car is a concrete class, consider the code fragment in Figure 3 (with lines numbered 1–5) and answer the following sub-questions:

```
1.  Car c1 = new Car();
2.  Car c2 = new Car();
3.  Car c3 = c1;
4.  c1 = null;
5.  c2 = c3;
```

Figure 3

【考点】Garbage Collection Mechanism

1) Indicate whether there are any objects eligible for garbage collection after executing the code in line 4 of Figure 3. Justify your answer. (3 Marks)

【Answer】Yes, c1 can be garbage collected.

【解析】

题目经过上述步骤之后是否有变量可以被垃圾回收。并进行判断。

Solution:

到第四步之前，经历了什么

新建 c1 实体（状态：c1 有），

新建 c2 实体（状态：c1 有，c2 有），

c3 指向了原 c1 实体的地址（状态：c1 有，c2 有，c3 为 c1）

c1 指空（状态：c1=null，c2 有，c3 为原 c1）

Yes. There is object eligible for garbage collection after executing the code in line4 of 3. c1 can be garbage collected now.

Finish. (3 Marks).

2) Now assume that the code in line 5 of Figure 3 has also been executed. Are there any objects eligible for garbage collection? Explain.(3 Marks)

到第五步之前，经历了什么

完成 4　c1 指空（状态：c1=null，c2 有，c3 为原 c1）

接下来 c2=c3（状态：c2 有是 4 之后的 c3，c3 为原 c1）

所以都指向原来的 c1，没办法清理。没被抛弃。

【Answer】No.

【解析】

题目经过上述步骤之后是否有变量可以被垃圾回收。并进行判断。

Solution:

No. There are no object eligible for garbage collection. Since c2 and c3 both pointed to something still exist.

Finish. (3 Marks)

【Review】可以被垃圾回收的对象 Objects eligible for GC

1、不再会使用的变量（指空的也算）

2、低内存自动清

c) The questions below are about Graphical User Interfaces (GUIs) in Java: (8 Marks)

【考点】GUI

i) What are the THREE main concepts when doing GUI programming in Java? Your answer must also include a brief description of each concept.(6 Marks)

【Answer】详见 Solution

【解析】

题目问编写 GUI 的三个主要概念并进行解释。死答案。

Solution:

Component, Container, Event.

Component: An objectthat the user can see on the screen and can also interact with.

Container: A component that can hold other components.

Event: An action triggered by the user (e.g.pressing a key, clicking a mouse button).

- There are *3 main concepts* when doing GUI programming in Java:
  - Component: An *object* that the user can see on the screen and can also interact with.
  - Container: A *component* that can hold other components.
  - Event: An *action* triggered by the user (e.g. pressing a key, clicking a mouse button).

Finish. (6 Marks, 2 for each concepts, 1+1.)

ii) Briefly describe how the THREE concepts indicated in part i) are used together when designing a GUI. (2 Marks)

【Answer】详见 Solution

【解析】

题目问编写 GUI 如何将它们使用在一起。意思就是三者在 GUI 中分别扮演什么样的角色。

Solution:

Designing a GUIinvolves creating **components**, putting them into **containers**, and arranging for the program to respond to **events**(e.g.by responding to mouse clicks).

Designing a GUI involves creating *components*, putting them into *containers*, and arranging for the program to respond to *events* (e.g. by responding to mouse clicks).

Finish. (2 Marks)

d) Consider the Java method in Figure 4 (with lines numbered 1–4) and describe how recursive algorithms are implemented in Java, using the code in Figure 4 as a reference example. Hint: You may want to refer to specific lines of code in your answer. (2 Marks)

```
1.  public int recurse(int x) {
2.    if (x == 0) { return x; }
3.    else { return x + recurse(x-1); }
4.  }
```

Figure 4

【Answer】详见 Solution

【考点】递归算法

【解析】

题目问 Java 中递归算法是如何实现的，结合上述代码进行阐释。

Solution:

Recursion is a technique in which a method calls itself a number of times to solve a problem.

Java 对于 Recursion 的 implements 分两部分：base case 和 recursion call

The case return x while x==0 is the base case.

else case return x+ recurse(x-1) is the recursion call.

Finish. (2 Marks)

Question 4: (25 Marks)

a) Answer the following questions about 2-dimensional arrays in Java. (4 Marks)

【考点】二维数组

i) Write the code to declare a 2-dimensional array named myStrings. This array should be capable of storing 10x10 String objects. (1 Mark)

【Answer】String[][] myStrings = new String[10][10];

【解析】

本题要求定义一个 10×10 的字符串类型二维数组。送分题。

Solution:

```
String[][] myStrings = new String[10][10];
```

注意格式：类型[][] 名称= new 类型[x][y]；要写清两个维度的大小。

ii) Imagine you wish to concatenate the String objects in each element of the myStrings array in part i). Write the Java code to concatenate all String objects into a single String object, and then print out the final results. (3 Marks)

Note: You are not required to write a complete Java program.

【Answer】详见 Solution

【考点】二维数组/字符串拼接算法

【解析】

本题要求写 java 代码拼接二维数组中的所有元素。核心有三步：第一步遍历二维数组元素，第二步依次拼接，第三步输出。不需要写出完整程序，那么只需要体现几个核心方法和变量即可。

Solution:

遍历肯定考虑循环，每循环到一个元素就拼接一次。此外不要忘记定义要输出的字符串变量。（拼接使用 concat 方法）

```
String result=""; //初始变量记得赋值！
for(int i=0;i<10;i++){
    for(int j=0;j<10;j++){
        result=result.concat(myStrings[i][j]);
    }
}
```

最后输出

```
System.out.println(result);
```

Final Answer:

```
String result="";
```

```
for(int i=0;i<10;i++){
    for(int j=0;j<10;j++){
        result=result.concat(myStrings[i][j]);
    }
}
System.out.println(result);
```

Finish. (3 Marks)

b) Answer the following questions about File Input/Output in Java:(10 Marks)

i) Whenever performing I/O in Java, it is necessary to use try and catch statements. Explain why this is. (3 Marks)

【Answer】详见 Solution

【考点】File I/O 的基本需求

【解析】

本题要求阐述文件输入输出必须使用 Try Catch 的原因。从 Try Catch 本身入手，显然这是处理异常所用的。文件输入输出必然有这种可能性。倾向从这个角度出发。

Solution:

File I/O can cause a large number of exceptions to be thrown, at least two, IOException and FileNotFoundException, which the exceptions are belongs to runtime error, not compile error, which are hard to predict. If we do not use try catch to find the exceptions, if any one exception triggers, the whole program will crash. (3 Marks)

个人认为：文件输出造成大量异常 1+属于 runtime 类不是 compile 类不好发现 1+如果不使用可能的后果 1

ii) Write a Java class called CountWriter. The class must generate a series of five sequential numbers (1, 2, 3, 4, 5) and write them to a file called count.txt. The class

must first create a new file, and then write the numbers into the file. Each number should be placed on a separate line. Figure 5 presents the data that the file count.txt should contain, after your code has executed.

```
1
2
3
4
5
```

Figure 5

Note: You must use a for loop to generate the numbers. The file should be created in the same directory as the program. If the file already exists, then the program must display the message "File already exists" and then terminate the program. (7 Marks)

【Answer】详见 Solution

【考点】File I/O 类的书写

【解析】

本题要求写一个文件读写 CountWriter 的类。创建文件，写入数字，还要一行一行写，代码执行完之后要求在文件中生成图 5 的样子。必须用循环，放在同一个文件夹内。文件如果存在需要显示已存在并终止程序。。我滴个妈。。要求好多。按照题目信息一句一句写。

Solution:

1、Write a Java class called CountWriter.

```
public class CountWriter{
}
```

2、The class must generate a series of five sequential numbers (1, 2, 3, 4, 5) and write them to a file called count.txt. You must use a for loop to generate the numbers.

生成数并读入文件。要求用循环。

```
String fileName="count.txt";
File file = new File(fileName);
FileWriter fileWriter= new FileWriter(file);
BufferedWriter bufferedWriter = new BufferedWriter(fileWriter);
for(int i=1;i<=5;i++){
    bufferedWriter.write(Integer.toString(i));
    bufferedWriter.write("\n");
}
bufferedWriter.close();
fileWriter.close();
```

最后别忘关闭文件哦～

如果要求放在同一个文件夹，fileName 这里不需要填地址，否则就需要文件位置。

3、If the file already exists, then the program must display the message "File already exists" and then terminate the program.

文件存在怎么写：

```
if(file.exists()){
    System.out.println("File already exists");
    System.exit(1);
}
```

4、此外还需要注意的点，io 库导入，文件输入输出放入 try/catch。

5、既然文件输入输出一定要考虑抛异常，文件输入输出部分进 try，后面 catch 要点明异常。

```
catch(IOException e){
    e.printStackTrace();
}
```

Final answer:

```
import java.io.*;
```

```
public class CountWriter{
    public static void main(String[] args){
        String fileName="count.txt";
        File file = new File(fileName);
        if(file.exists()){
            System.out.println("File already exists");
            System.exit(1);
        }
        try{
            FileWriter fileWriter= new FileWriter(file);
            BufferedWriter bufferedWriter = new BufferedWriter(fileWriter);
            for(int i=1;i<=5;i++){
                bufferedWriter.write(Integer.toString(i));
                bufferedWriter.write("\n");
            }
            bufferedWriter.close();
            fileWriter.close();
        }
        catch(IOException e){
            e.printStackTrace();
        }
    }
}
```

Finish. (7 Marks)

c) Write the Java code for a simple GUI. This should include a JButton that is placed inside of a JFrame. The JButton should contain the text "Hello!". You must also add event handling capabilities to your code: when a user clicks the button, the text in the button should change to "Bye Bye!". (5 Marks)

Note: You are not required to write a complete Java program.

【Answer】详见 Solution

【考点】简单 GUI 的书写

【解析】

本题要求写一个 GUI 文件。要求包括一个按钮，先是显示 Hello，之后要求用户点击按钮之后，按钮的字改变成 Bye Bye！按照题目信息一句一句写。

Solution:

1、导入相关库

有 JButton，JFrame 以及事件。所以需要如下两个库：

```
import javax.swing.*;
import java.awt.event.*;
```

2、搭框架

```
JFrame frame= new JFrame();
frame.setVisible(true);
```

3、放按钮

在显示 frame 之前

```
JButton button = new JButton("Hello!");
button.addActionListener(this);
frame.getContentPane().add(button);
```

4、设定事件

```
button.addActionListener(this);
```

类带上 implements ActionListener

还有方法：

```
public void actionPerformed(ActionEvent e){
    button.setText("Bye Bye!");
}
```

Final Answer:

```
import javax.swing.*;
```

```
import java.awt.event.*;
public class simpleGUI implements ActionListener{
    JButton button;
    public static void main(String[] args){
        simpleGUI myGUI = new simpleGUI(); //不必要
        myGUI.go(); //不必要
    }
    public void go(){
        JFrame frame= new JFrame();
        button = new JButton("Hello!");
        button.addActionListener(this);
        frame.getContentPane().add(button);
        frame.setSize(100,100); //不必要
        frame.setVisible(true);
    }
    public void actionPerformed(ActionEvent e){
        button.setText("Bye Bye!");
    }
}
```

Finish. (5 Marks)

（不用写成代码形式，上文提到关键点抓住即可。）

d) The questions below refer to exceptions and assertions in Java: (6 Marks)

i) What is the output of the program in Figure 6 (with lines numbered 1-13) when assertions are disabled and when assertions are enabled? Justify your answers.(4 Marks)

```
public class AssertionTest {
  public static void main(String[] args) {
    try {
      assert(false);
    }
    catch (Exception ex) {
      System.out.println("Caught an exception!");
    }
    catch (Error err) {
      System.out.println("Caught an error!");
    }
  }
}
```

Figure 6

【Answer】详见 Solution

【考点】Assertion

【解析】

本题给出代码，要求分别写出 assertion 忽略和不忽略情况下代码输出的不同。正常情况下 assertion 是忽略的，除非给指令-ea。因为用途就是检验代码正误使用，正常情况下不会用。根据这个用途所以正常情况下 assertion 忽略。

Solution:

Disable assertion: No output.

Since the assertion is ignored, nothing exception happens, so no output.

Enable assertion: Caught an error!

Since assertion is false, then program causing an error exception, then this exception is caught and print out “Caught an error!”.

Finish. (4 Marks)

ii) Write a new Exception class called LargeDataException. This must be defined in a class file. You must also state the name of the file that you would write the code into. (2 Marks)

Note: You only need to write the class declaration.

【**Answer**】public class LargeDataException extends Exception

【考点】自定义异常

【解析】

本题要求只写一个类的声明，自定义一个异常。自定义异常我们想到什么？两个关键点。

Solution:

要求写自定义异常，只要类的声明。类名=异常名，之后 extends Exception

public class **LargeDataException** extends **Exception**.